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# ★ Alpha Examples for Beginners

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## Momentum after news

### Hypothesis

After news is released, if a stock takes a longer time to rise, it may show strong evidence of upward momentum, and it could be beneficial to take a long position in it.

### Implementation

Use the fundamental data field 'nws12\_prez\_4l' to capture this hypothesis. Backfill it with two years of data to ensure there is no drop in coverage.

### Hint to improve the Alpha

Can increasing weight on more liquid stocks (Stocks with high volume) help pass the sub-universe test?

```
1 | ts_backfill(vec_avg(nws12_prez_4l), 504)
```

[Open example alpha in Simulate](#)

### Simulation Settings

| Region | Universe | Language        | Decay | Delay | Truncation | Neutralization | Pasteurization | NaN Handling | Unit Handling |
|--------|----------|-----------------|-------|-------|------------|----------------|----------------|--------------|---------------|
| USA    | TOP500   | Fast Expression | 0     | 1     | 0.08       | Industry       | On             | Off          | Verify        |

Hypothesis

Pretax income is a good measure of a company's financial health and profitability. Companies with increasing pretax income may have good growth prospects and higher potential for increasing stock price. Thus, you tend to long stocks with increasing pretax income and short stocks with decreasing pretax income.

Implementation

Use the time-series rank operator to compare the trend of the fundamental data field pretax income over the past 2 years, and use the quantile operator to normalize the result.

Hints to Implement

Boost the signal with sales data. For example, if the company has higher sales, it is more likely to outperform.

```
1 | quantile(ts_rank(pretax_income,250))
```

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Simulation Settings

| Region | Universe | Language        | Decay | Delay | Truncation | Neutralization | Pasteurization | NaN Handling | Unit Handling |
|--------|----------|-----------------|-------|-------|------------|----------------|----------------|--------------|---------------|
| USA    | TOP3000  | Fast Expression | 4     | 1     | 0.01       | Market         | On             | Off          | Verify        |

Operating Earnings Yield

Hypothesis

If the operating income of a company is currently higher than its past 1 year history, buy the company's stock and vice-versa.

Implementation

Using ts\_rank to identify current performance of the company compared to its own history, using the fundamental data field "operating\_income".

Hints to Implement

Rather than comparing the value directly, can calculating a ratio that includes stock market moves, improve the signal?

```
1 | ts_rank(operating_income.252)
```

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Simulation Settings

| Region | Universe | Language        | Decay | Delay | Truncation | Neutralization | Pasteurization | NaN Handling | Unit Handling |
|--------|----------|-----------------|-------|-------|------------|----------------|----------------|--------------|---------------|
| USA    | TOP3000  | Fast Expression | 0     | 1     | 0.08       | Subindustry    | On             | Off          | Verify        |

# Appreciation of liabilities

## Hypothesis

An increase in the fair value of liabilities could indicate a higher cost than expected. This may deteriorate the company's financial health, potentially leading to lower profitability or financial distress.

## Implementation

Go short when there is an increase in the fair value of liabilities within a year and long when the opposite occurs using fundamental data.

## Hints to Implement

Could observing the increase over a shorter period improve accuracy?

```
1 | -ts_rank(fn_liab_fair_val_l1_a,252)
```

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Simulation Settings

| Region | Universe | Language        | Decay | Delay | Truncation | Neutralization | Pasteurization | NaN Handling | Unit Handling |
|--------|----------|-----------------|-------|-------|------------|----------------|----------------|--------------|---------------|
| USA    | TOP3000  | Fast Expression | 0     | 1     | 0.08       | Subindustry    | On             | Off          | Verify        |

# Deferred Revenue

## Hypothesis

Firms with high deferred revenue will surprise the market in the future when the deferred revenue is recognized.

## Implementation

fnd6\_drc field refers to deferred revenue. To improve the coverage of the datafield, ts\_backfill operator is applied. Deferred revenue is divided by total assets to account for the size of a firm.

Instead of relying on the raw value of the ratio to decide weights of stocks, use cross-sectional operators. Utilize group based operators to compare across similar stocks.

```
1 | ts_backfill(fnd6_drc, 252)/assets
```

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Simulation Settings

| Region | Universe | Language        | Decay | Delay | Truncation | Neutralization | Pasteurization | NaN Handling | Unit Handling |
|--------|----------|-----------------|-------|-------|------------|----------------|----------------|--------------|---------------|
| USA    | TOP3000  | Fast Expression | 0     | 1     | 1          | Sector         | On             | Off          | Verify        |

# Reducing debt

**Hypothesis:** Take a long position in companies whose debt has decreased compared to the past, and conversely, take a short position in companies whose debt has increased.

**Implementation:** Use the fundamental data 'debt' to capture this hypothesis. Use the time-series quantile operator to compare the trend of the fundamental data field over the past six months.

**Hint to improve the Alpha:** Utilize the operator's driver parameter to transform it into a different distribution.

```
1 | -ts_quantile(debt, 126)
```

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Simulation Settings

| Region | Universe | Language        | Decay | Delay | Truncation | Neutralization | Pasteurization | NaN Handling | Unit Handling |
|--------|----------|-----------------|-------|-------|------------|----------------|----------------|--------------|---------------|
| USA    | TOP3000  | Fast Expression | 0     | 1     | 0.01       | Market         | On             | Off          | Verify        |

# Power of leverage

**Hypothesis:** Companies with high liability-to-asset ratios – excluding those with poor financial health or weak cashflows – often leverage debt as a strategic tool to pursue aggressive growth

...earnings, as they reflect better the capital's high potential opportunities.

**Implementation:** Use the 'liabilities' and 'assets' to design the ratio.

**Hint to improve the Alpha:** This ratio can vary significantly across industries. Would it be worth considering alternative neutralization settings?

1 | liabilities/assets

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Simulation Settings

| Region | Universe | Language        | Decay | Delay | Truncation | Neutralization | Pasteurization | NaN Handling | Unit Handling |
|--------|----------|-----------------|-------|-------|------------|----------------|----------------|--------------|---------------|
| USA    | TOP3000  | Fast Expression | 0     | 1     | 0.01       | Market         | On             | Off          | Verify        |

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